Kernel Regression Methods

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Introduction

In this homework we implemented **Kernelized Ridge Regression (KRR)** and **Support Vector Regression (SVR)** with a polynomial and an RBF kernel each. KRR minimizes a regularized squared loss in the dual space. SVR minimizes an ε-insensitive loss via quadratic programming [1]. We applied both to a 1D sine dataset for visual inspection and to a multidimensional housing dataset for MSE evaluation via cross-validation.

Part 1: Sine Dataset

The polynomial kernel of degree M is:

$$\kappa(x,x') = (1 + x \cdot x')^M \tag{1}$$

Expanding the binomial yields an implicit polynomial feature map up to degree M.

The RBF kernel with bandwidth σ is:

$$\kappa(x,x') = \exp\left(-\frac{\|x-x'\|^2}{2\sigma^2}\right) \tag{2}$$

Its value decays with squared distance into an infinite-dimensional implicit feature space.

Both methods apply the kernel trick, replacing inner products with kernel evaluations. **KRR** finds $\alpha = (K+\lambda I)^{-1}y$ where $K_{ij} = \kappa(x_i, x_j)$, and predicts $f(x) = \sum_i \alpha_i \kappa(x_i, x)$. **SVR** adds an ε-insensitive tube: residuals smaller than ε incur no penalty. The dual QP uses K as its quadratic term, with decision variables $[\alpha_1, \alpha_1^*, \alpha_2, \alpha_2^*, \ldots]$ and box constraint $C = 1/\lambda$. Predictions are $f(x) = \sum_i (\alpha_i - \alpha_i^*)\kappa(x_i, x) + b$, where only points outside the tube (support vectors) have $\alpha_i > 0$ or $\alpha_i^* > 0$.

We applied both methods to `sine.csv` (200 noisy observations of a sinusoidal signal on $[0,20]$). Parameters were chosen manually to obtain a good fit and a sparse SVR solution. Results are in Table 1.

RBF fits the periodic signal well (train MSE ≈ 0.07) and most predictions land inside the ε-tube, leaving only 85/200 points as support vectors. A polynomial of degree 5 cannot replicate a multi-period sine regardless of parameters, so residuals are large relative to ε and 141/200 points become support vectors. The high SV count for polynomial SVR is a direct consequence of kernel mismatch, not a tuning failure. Figure 1 shows all four fits.

Method	Kernel	σ/M	λ	ε	Train MSE
KRR	RBF	σ=1.0	0.01	–	**0.069**
KRR	Polynomial	M=5	1.0	–	0.977
SVR	RBF	σ=1.0	0.01	0.3	**0.072**
SVR	Polynomial	M=5	1.0	0.3	1.020

Table 1. Sine dataset results. SVR with RBF produced 85/200 support vectors and SVR with Polynomial produced 141/200.

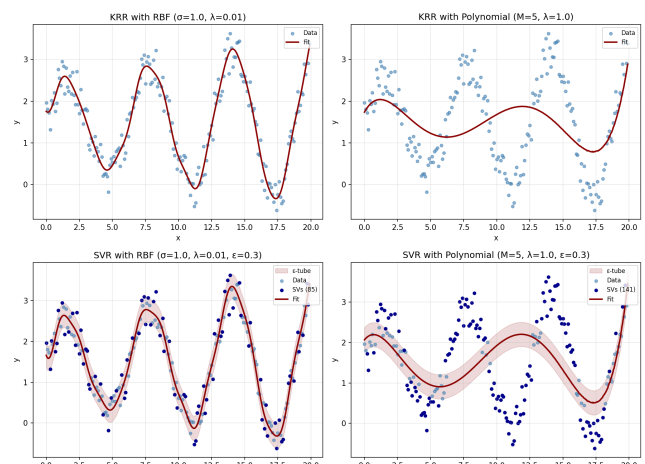

Figure 1. Sine dataset fits. Top row: KRR with RBF (left) and Polynomial (right). Bottom row: SVR with RBF (left) and Polynomial (right). Dark blue points are support vectors and the red shaded band is the ε-tube.

Part 2: Housing Dataset

We applied both methods to `housing2r.csv` (300 samples, 5 features). Features were standardized before computing kernel distances. We measured MSE via 5-fold CV across a grid of kernel parameter values, comparing fixed $\lambda = 1$ against the best λ from CV over $\{0.001, 0.01, 0.1, 1, 10, 100\}$. For SVR we used $\varepsilon = 2.0$ (roughly 20% of the target standard deviation), giving sparse solutions with 75-144 support vectors out of 300. Figure 2 shows results for all four method-kernel combinations, with support vector counts on a secondary axis for SVR.

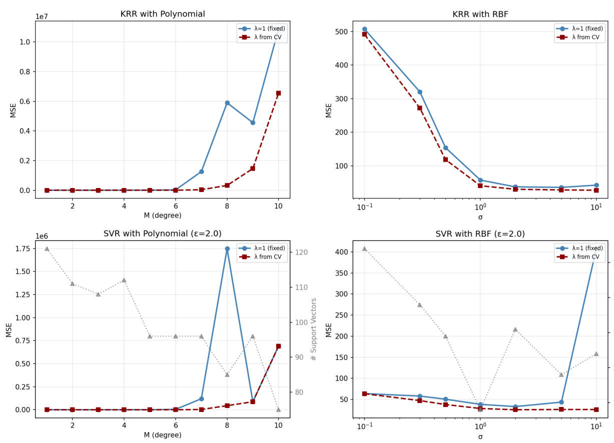

Figure 2. Housing dataset: CV MSE vs kernel parameter for KRR (top) and SVR (bottom) with polynomial (left) and RBF (right) kernels. Dashed lines show λ selected by inner CV and solid lines use fixed $\lambda = 1$. Gray triangles (secondary axis) show support vector counts for SVR.

Table 2 shows the best configurations. For KRR, RBF outperforms polynomial. For SVR, the best polynomial ($M = 2$) matches RBF almost exactly. Polynomial kernels become numerically unstable for $M \geq 4$ even after standardization, so MSE grows sharply at higher degrees despite heavy regularization. RBF peaks around $\sigma \in [2, 10]$, near the typical inter-point distance after standardization.

Method	Kernel	Best param	CV MSE
KRR	Polynomial	$M = 2$	27.63
KRR	RBF	$\sigma = 10$	**27.13**
SVR	Polynomial	$M = 2$	25.65
SVR	RBF	$\sigma = 2$	**25.64**

Table 2. Best CV MSE for each method and kernel on the housing dataset.

Discussion

Similarities. Both methods use the same kernel trick and produce identical predictions in the limit $\lambda \to 0$, $\varepsilon \to 0$. Larger λ yields a smoother, more biased fit for both.

Differences. SVR achieves slightly lower MSE (≈ 25.6 vs ≈ 27.1 at best). The ε-insensitive loss discards small residuals, making SVR more robust to noise. SVR's solution is also sparse: only support vectors contribute to predictions, while KRR uses all training points. KRR solves one linear system ($O(n^3)$), while SVR requires a quadratic program and is considerably slower. SVR also adds a hyperparameter ε more sensitive to mistuning than λ.

Preference. For this dataset, KRR is the more practical choice: the MSE gap is small (≈ 2 points), it is faster to fit and has one fewer hyperparameter. SVR's sparsity would matter at larger scale or when inference cost is a concern, but at $n = 300$ neither applies.

References

[1] Alex J. Smola and Bernhard Schölkopf. A tutorial on support vector regression. *Statistics and Computing*, 14(3):199–222, 2004.